

On the Recurrence of $2/3$: A Reply to Austin's Algebraic Note

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in reply to Peter Austin, *A Purely Algebraic Short Note on the Repeated Appearance of $2/3$* (May 2026)

for the IPI correspondence group — May 2026

.1 Purpose and stance

Peter, thank you for the note. It is careful, the algebra is correct, and it reconstructs the FFGFT lepton sector accurately from the public summaries. What follows is not a defence and not a request for validation: it is the set of internal derivations your note effectively asks for. Where your reconstruction stops at the public numbers, FFGFT already carries the structure one level deeper, and in two places your note independently arrives at conclusions that the framework reached on its own. I will use your section numbering throughout.

For the record, the canonical charged-lepton coefficients are

$$(r_e, r_\mu, r_\tau) = \left(\frac{4}{3}, \frac{16}{5}, \frac{25}{9}\right), \quad (p_e, p_\mu, p_\tau) = \left(\frac{3}{2}, 1, \frac{2}{3}\right),$$

with $m_i = r_i \xi^{p_i} v$ and $\xi = 4/30000$. Your note uses exactly these; they are current.

.2 The ϵ -deformation discriminator (your §6)

I accept the discriminator without reservation. Coincidence at $D = 3$ is not identity of mechanism, and the continuation $D = 3 + \epsilon$ is the right instrument to separate the roads. This is worth stating plainly because FFGFT reaches the *same* separation from the inside, by a different route (see §5 below): the exponent-ladder value of the Koide quantity and the equal-projector condition $A = B$ do not coincide exactly — they meet *near* $2/3$ and part company, precisely as your ϵ -analysis predicts. We are not in disagreement on the method; we agree on the conclusion.

.3 The ladder is geometric, not arithmetic (your §11)

Your §11 is correct that $(\frac{3}{2}, 1, \frac{2}{3})$ is not a uniform $1/3$ ladder: the gaps are

$$p_e - p_\mu = \frac{1}{2}, \quad p_\mu - p_\tau = \frac{1}{3}.$$

The resolution is that the ladder is not arithmetic at all. It is a *geometric* sequence with ratio $q = 2/3$:

$$p_{n+1} = \frac{2}{3} p_n, \quad \frac{3}{2} \xrightarrow{\times 2/3} 1 \xrightarrow{\times 2/3} \frac{2}{3}.$$

The unequal differences $1/2$ and $1/3$ are then *consequences* of a constant ratio, not independent rules to be posted. So the electron does not need its own ad-hoc rule; it is the base rung of the sequence. Two equivalent readings of the same fact:

- **Ratio reading.** $q = 2/3$ is the inverse of the tetrahedral geometry factor $3/2$ (four vertices over three edges per vertex). The same number that orders the ladder is, numerically, the Koide value — which is exactly why the two appear linked.
- **Complement reading (your road).** At the $\mu \rightarrow \tau$ rung the step is $1 - 1/D = 2/3$ with $D = 3$, so $p_\tau = 1 - 1/D$. This places FFGFT on your “complement of step” road, as you correctly identified, and *not* on the equal-projector road $2/D$.

In the harmonic picture the rungs sit on a logarithmic torus in steps of a musical third, $\Delta p = 1/3$. The electron at $p_e = 3/2 = 1 + 1/(D-1)$ is a *half-integer* rung — a geometric midpoint between winding levels (the spin-winding contribution), not a full winding step. That is the structural origin of the $1/2$ gap.

.4 Topological $D = 3$ versus the fractal dimension (your §11–§12)

Your §12 asks whether the $1/3$ step is tied to the integer $D = 3$ or to $D_f = 3 - \xi$, the latter giving a downward shift $2/3 - \xi/9$. The framework's answer is specific, and it dissolves the worry for the Koide case in particular.

First, there are *two* derived fractal dimensions, not one:

$$D_f^{\text{space}} = 3 - \xi \approx 2.9998667 \quad (\text{local metric; GRB time-of-flight}),$$

$$D_f^{\text{eff}} \approx 2.973 \quad (\text{cumulative scale flow, Planck to electroweak}).$$

Both are *derived*, not assumed. The exponent step is tied to the *integer, topological* torus dimension $D = 3$; the metric $3 - \xi$ enters elsewhere (dispersion), and the effective 2.973 enters only the absolute normalisation through $K_{\text{frak}} = (D_f^{\text{eff}}/3)^{D_f^{\text{eff}}/2} = 1 - 100\xi$.

Why no $\xi/9$ correction reaches Koide

The Koide quantity Q is a pure mass *ratio*. Every scale-dependent factor — the vacuum value v , the fractal correction K_{frak} , and any D_f -level shift — is common to all three leptons and cancels in Q . Fractal corrections attach to *absolute* scales, never to ratios. Your §12 shift $2/3 - \xi/9$ would be a real effect on an absolute prediction; it simply does not enter Q , because Q never sees the absolute scale.

So for Koide the choice is settled in favour of your option (a): the step is on topological $D = 3$, and D_f does not perturb the ratio.

.5 Forced or merely fitted? An honest answer (your §10, §13)

This is the central question of your note, and FFGFT’s internal answer is already on record — and it is more interesting than a clean “forced.”

What is structurally forced

The mass *ratios* are not fitted. Two points:

- m_μ/m_e follows two independent ways that agree: geometrically from the ladder, $\frac{12}{5}\xi^{-1/2} \approx 207.8$, and fractally from $(D_f^{\text{eff}}/3)^{D_f^{\text{eff}}/2} \cdot f(\xi) \approx 206.8$. The agreement fixes K_{frak} uniquely — a consistency test, not a fit.
- In the compact representation the coefficients are not free rationals: $\frac{2}{3} = 3\sqrt{3}/(2\pi\alpha^{1/2})$ and $\frac{8}{5} = 9/(4\pi\alpha)$, built so that α cancels exactly. The rationals are α -geometric invariants.

What the ladder does not force

It does not force the equal-projector condition $A = B$ exactly. Inserting the FFGFT masses into the Koide formula gives

$$Q_{\text{FFGFT}} \approx 0.6677, \quad \Delta_K = 3Q_{\text{FFGFT}} - 2 \approx +0.003,$$

i.e. your residual diagnostic $\Delta_K = B/A - 1$ is small but nonzero: $A = B$ holds to about 0.16% in Q , not exactly. This is deliberate and honest. The reason is structural, and it is your own point in disguise: the Koide denominator $(\sum \sqrt{m_i})^2$ generates cross-terms with exponents

$$\xi^{5/4}, \quad \xi^{13/12}, \quad \xi^{5/6},$$

none of which is a multiple of $1/3$. They lie *outside* the FFGFT structure space $\{p\} \subset \frac{1}{3}\mathbb{Z}$; they are artefacts of the square-root composition, not torus modes. Exact $Q = 2/3$ would require

$$m_e + m_\mu + m_\tau = 4(\sqrt{m_e m_\mu} + \sqrt{m_e m_\tau} + \sqrt{m_\mu m_\tau}),$$

which holds not at $\xi = 4/30000$ but at $\xi_{\text{Koide}} \approx 1.372 \times 10^{-4}$, about 2.9% larger. In other words, the equal-projector condition is an algebraic side-condition that is *independent* of the torus geometry — which is exactly what your ϵ -deformation says it should be: the ladder road and the equal-projector road are different mechanisms that meet near $2/3$ and separate.

The honest position

The ladder *predicts* Q to 0.16% from (r_i, p_i, ξ) with no Koide input. The exact experimental $Q = 2/3$ then *confirms* the exponent structure from an independent direction; it is neither fitted into FFGFT nor algebraically forced by it. The small residual is real and located precisely where your cross-term analysis predicts.

.6 Three roads and the map (your §13)

I agree with your framing: the Koide projector form, the FFGFT ξ -ladder, and Paul's 24-cell rank split are three different primitive objects, and a relation between them must be exhibited by a map, not by a shared numeral. The FFGFT-to-Koide map is explicit,

$$(r_i, p_i, \xi) \mapsto x_i = \sqrt{r_i} \xi^{p_i/2} \sqrt{v} \mapsto (A, B) \mapsto Q,$$

and your trigonometry-free target is, in our notation, identical to the generated quantity

$$Q_{\text{FFGFT}} = \frac{\sum_i r_i \xi^{p_i}}{(\sum_i \sqrt{r_i} \xi^{p_i/2})^2},$$

since $\sqrt{v}$ cancels. We are computing the same object. The point of disagreement, if any, is not the map but the claim of exactness: the map sends the ladder to $Q \approx 0.6677$, not to $2/3$ on the nose, for the reason in §5.

.7 Closing

So, briefly: your method is right; your placement of FFGFT on the complement road is right; the ladder asymmetry resolves as a geometric sequence of ratio $2/3$; the D_f shift does not reach Koide because Q is a ratio; and the “forced versus fitted” question has the honest answer that the ladder predicts Q to 0.16% and the exact value is a confirmation along an independent road — which your own ϵ -analysis anticipates. If it is useful, you are welcome to post this alongside your note; I think the two together make the cleaner statement.

— J.P.